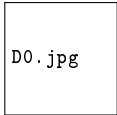


Lecture 23: SQL DML

INSERT, UPDATE, and DELETE

Z2004: Database Management Systems



D0.jpg

IIT Madras Zanzibar

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What is DML?

DML (Data Manipulation Language) is the part of SQL used to **change the data** stored in tables.

Core DML commands:

- ▶ **INSERT** — add new rows
- ▶ **UPDATE** — modify existing rows
- ▶ **DELETE** — remove rows

Note: **SELECT** reads data (often called DQL), not DML.

DML Syntax Patterns

INSERT (new row):

```
INSERT INTO TableName (col1, col2, ...)
VALUES (val1, val2, ...);
```

UPDATE (change rows):

```
UPDATE TableName
SET      col1 = newValue,
         col2 = expression
WHERE    condition;
```

DELETE (remove rows):

```
DELETE FROM TableName
WHERE    condition;
```

Workflow: write WHERE first → run as SELECT → then run UPDATE/DELETE.

INSERT (Patterns) — Common INSERT Forms

```
-- Insert one row
INSERT INTO table_name (column1, column2)
VALUES (value1, value2);

-- Insert multiple rows
INSERT INTO table_name (column1, column2) VALUES
(value1a, value2a),
(value1b, value2b),
(value1c, value2c);

-- Insert from another table
INSERT INTO table2
SELECT * FROM table1 WHERE condition;
```

INSERT INTO — Adding Rows

```
USE SchoolDB;

-- 1) Insert a student (SID is IDENTITY, so we omit it)
INSERT INTO Student (Name, Dept, GPA)
VALUES (N'Asha Ali', 'CSE ', 3.80);

-- 2) Insert a course (natural key CID)
INSERT INTO Course (CID, CName, Credits)
VALUES ('DB101', N'Introduction to Databases', 3);

-- 3) Enrol the student (FKs must match existing Student and Course)
INSERT INTO Enrolment (SID, CID, Grade)
VALUES (1, 'DB101', NULL);
```

Tip: Always name columns in INSERT.

UPDATE — Modifying Existing Rows

```
USE SchoolDB;

-- 1) Update one student's GPA
UPDATE Student
SET     GPA = 3.90
WHERE   SID = 1;

-- 2) Update multiple columns at once
UPDATE Student
SET     Dept = 'CSE ',
        GPA  = GPA + 0.10
WHERE   SID = 1;

-- Preview rows BEFORE updating
SELECT SID, Name, Dept, GPA
FROM   Student
WHERE  SID = 1;
```

Rule: write WHERE first, test it as SELECT, then run UPDATE.

DELETE — Removing Rows

```
USE SchoolDB;

-- Preview what you are about to delete
SELECT SID, Name, Dept
FROM Student
WHERE Dept = 'CSE';

-- Delete rows by condition
DELETE FROM Student
WHERE Dept = 'CSE';

-- Delete child rows (Enrolments) for a given course
DELETE FROM Enrolment
WHERE CID = 'DB101';
```

Warning: Never run DELETE without WHERE.

DML Commands Compared — DELETE vs TRUNCATE vs DROP

Dimension	DELETE	TRUNCATE	DROP
Removes rows	Yes (selective)	All rows	All rows
Removes structure	No	No	Yes
WHERE clause	Yes	No	No
Triggers fire	Yes	No	No
Can rollback	Yes (if in txn)	Yes (logged)	No*
Resets IDENTITY	No	Yes	Removed
Speed	Slow (row-by-row)	Fast	Instant
Referential checks	Yes	Yes (FKs block)	Yes

*Only if inside an explicit transaction block

Common DML Mistakes

Mistake 1: UPDATE without WHERE

```
-- DANGER: updates every row!
```

```
UPDATE Student  
SET      Dept = 'CSE  ';
```

```
-- Safer pattern
```

```
UPDATE Student  
SET      Dept = 'CSE  '  
WHERE    SID = 1;
```

Mistake 2: DELETE without WHERE

```
-- DANGER: empties the table!
```

```
DELETE FROM Enrolment;
```

```
-- Correct
```

```
DELETE FROM Enrolment  
WHERE    SID = 1 AND CID = 'DB101';
```

Mistake 3: Foreign key violation

```
-- Fails if Student (SID=999) doesn't exist
```

```
INSERT INTO Enrolment (SID, CID,  
                      Grade)  
VALUES (999, 'DB101', 'A');
```

```
-- Fix: insert the parent first
```

```
INSERT INTO Student (Name, Dept,  
                    GPA)  
VALUES (N'Test', 'CSE  ', 3.00);
```

Best practice: test your WHERE as a SELECT first.

Key takeaways (L23: SQL DML)

- ▶ INSERT adds new rows; prefer naming the column list explicitly
- ▶ UPDATE modifies existing rows; always include a correct WHERE
- ▶ DELETE removes rows; always include a correct WHERE
- ▶ Safe workflow: test WHERE as a SELECT first, then run UPDATE/DELETE